MAJOR PROJECT BLUEPRINT & COMPLETE SYSTEM SPECIFICATION

Autonomous Multi-Agent EV Battery Life & Range Predictor

Analyzing Real-World Range Decay (650 km → 550 km) Across NMC, LFP, Sodium-ion & Solid-State Batteries via Open-Source AI Agents & Machine Learning

Comprehensive System Blueprint: This master document integrates the end-to-end multi-agent architecture, the 5-team modular workload distribution, raw datasets, machine learning models, autonomous agent frameworks, software stack dependencies, mathematical governing equations, reference scientific literature, and the 6-week execution roadmap.

1. End-to-End Multi-Agent Architecture

The pipeline decouples numerical transformations from reasoning. Heavy matrix calculations and continuous feature extraction execute within deterministic Python/C++ binaries, while autonomous LLM agents supervise data hygiene, execute scenario simulations, and synthesize materials science literature.

2. Team Deliverables & Modular Workload Breakdown (5 Roles)

MEMBER 1

Telemetry Ingestion & Preprocessing

Collects and standardizes laboratory aging cycles across NMC, LFP, and Sodium-ion cells. Applies Savitzky-Golay and moving-average filters to remove ADC voltage noise, segregates constant-current (CC) charge from dynamic discharge, interpolates non-uniform timestamps (Δt = 1s), and computes baseline SOH.

> **Deliverable Contract:** Cleaned `.parquet` files with schema: `cycle_id, time_sec, voltage_v, current_a, temp_c, capacity_ah, soh_pct`.

MEMBER 2

Feature Engineering & ML Predictor

Computes electrochemistry markers: Incremental Capacity Analysis (***dQ/dV***) peak tracking, Differential Voltage (***dV/dQ***), thermal stress hours outside 20°C–30°C, and high C-rate exposure. Trains LightGBM tabular regressors and an LSTM sequence model to forecast multi-cycle capacity fade.

> **Deliverable Contract:** Exported inference models (`model.pkl` / ONNX) taking feature vectors and returning predicted future SOH and RUL.

MEMBER 3

Real-World Range Derivation Engine

Translates nominal lab degradation into physical kilometers. Implements road load equations (aerodynamic drag, rolling resistance, inertial acceleration) and HVAC cabin thermal loads to demonstrate why factory-rated 650 km packs drop to 550 km or lower under hot/cold ambient conditions and aggressive driving.

> **Deliverable Contract:** Python API (`RangeCalculator`) taking vehicle telemetry, SOH, and ambient weather to return instant real-world range.

MEMBER 4

Multi-Agent Simulation & BMS Advisory

Builds a multi-agent sandbox using LangGraph or AutoGen. Pits an aggressive *Driver Persona Agent* (demanding rapid DCFC at high ambient temperatures) against a *BMS Safety Supervisor Agent* enforcing thermal cutoffs and C-rate ceilings. Derives optimal charging bounds (20%–80% SOC rules) to maximize lifespan.

> **Deliverable Contract:** Multi-agent orchestration script yielding simulated multi-year lifecycle comparisons and custom battery preservation guidelines.

MEMBER 5

System Integrator, Literature Synthesis (RAG) & Streamlit Dashboard Lead

Constructs an open-access RAG agent querying ArXiv/ScienceDirect literature to synthesize next-generation Solid-State (sulfide/oxide electrolytes, dendrite suppression) and Sodium-ion cathode designs. Integrates modules from Members 1–4 into a reactive Streamlit dashboard with full controls for chemistry, climate, and driving habits.

> **Deliverable Contract:** Deployable GitHub repository with containerized Docker configuration, interactive web UI, and finalized project report graphics.

3. Prerequisite Datasets

Battery Chemistry	Recommended Dataset Source	Key Features Extracted	Role in System
Lithium-ion (NMC)	NASA Ames Prognostics Center of Excellence (Cells B0005, B0006, B0018)	Voltage, Current, Cell Temp, Cycle Time, Capacity (Ah), EIS Impedance	Benchmark capacity fade, SEI layer growth, and cycle-life baseline.
Lithium Iron Phosphate (LFP)	BatteryArchive.org (Sandia National Lab & HNEI Cell Aging Sets)	Multi-temperature cycling (15°C, 25°C, 35°C), DoD, high C-rate cycling	Evaluating flat voltage plateau behavior and high thermal resilience.
Sodium-ion (Na-ion)	IonTech Open Repositories / Synthetic Cycling via PyBaMM	Low-temperature discharge curves (−20°C to 25°C), rate capability	Benchmarking low-cost alternative chemistry and sub-zero retention.
Solid-State Battery	ArXiv Literature Extracts / PyBaMM Degradation Synthesis	Interface impedance, critical current density, pressure-voltage curves	Simulating next-generation dendrite-free lifespan trajectories.
Vehicle Dynamics	EPA / WLTP Standard Drive Cycles (UDDS, HWFET) & Open CAN logs	Velocity profiles (km/h), road inclination (θ), auxiliary HVAC kW load	Translating pack SOH into real driving distance (650 km → 550 km).

4. Machine Learning & Computational Models

A. Electrochemistry Feature Extraction Tools

PyBaMM (Python Battery Mathematical Modeling): Physics-based solver implementing Doyle-Fuller-Newman (DFN) and Single Particle Models (SPM) to generate baseline synthetic curves for Solid-State and Sodium-ion cells where public cycling is scarce.

Scipy Signal Processing: Numerical filters (Savitzky-Golay) to smooth discrete raw voltage measurements, enabling robust second-derivative extraction for Incremental Capacity (dQ/dV) and Differential Voltage (dV/dQ).

B. Supervised Degradation & SOH Predictors

LightGBM / XGBoost Regressors: Primary gradient-boosted decision trees trained on cycle-level aggregated tabular metrics (mean temperature, maximum C-rate, time spent above 80% SOC). Delivers high predictive accuracy (MAE < 1.5%) with microsecond inference times.

PyTorch Sequence Model (LSTM / Bi-LSTM): Recurrent neural network that ingests sliding historical windows (e.g., past 30 cycles) of multivariate time-series to capture long-term temporal dependencies and non-linear degradation inflection points (knee points).

C. Physical Vehicle Road Load Model

Lumped-Parameter Energy Equation: Computes instantaneous mechanical power at the wheels:

$$P_{ ext{wheel}} = \left(m \cdot a + rac{1}{2} ho C_d A v^2 + m g C_{rr} \cos heta + m g \sin heta ight) v + P_{ ext{HVAC}}(T_{ ext{ambient}})$$

Coupled with pack degradation to model the practical drop from 650 km factory rating down to 550 km on the road.

5. AI Agent Frameworks & Zero-Cost Inference

- **LangGraph (Open Source):** State-machine orchestration engine managing sequential and cyclical transitions across data cleaning nodes, ML feature generators, and advisory report agents.
- **Microsoft AutoGen:** Autonomous multi-persona simulation framework. Enables simulated discourse between an aggressive driving profile and a virtual BMS supervisor to identify optimal charging bounds.
- **Google AI Studio (Gemini 1.5 Flash API):** Provides free-tier cloud inference for multi-agent reasoning, data audit verification, and scientific synthesis with generous rate limits.
- **ChromaDB / FAISS:** Embedded local vector store for document indexing, chunking, and semantic retrieval over battery research papers.

6. Software Stack & Environment Dependencies

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# 1. Core Data Science, Electrochemistry & Machine Learning pip install numpy pandas scipy scikit-learn xgboost lightgbm torch pybamm # 2. Multi-Agent Systems, RAG & LLM Orchestration pip install langgraph langchain langchain-google-genai autogen-agentchat chromadb sentence-transformers # 3. Interactive Web Dashboard, Visualization & Utilities pip install streamlit plotly matplotlib seaborn python-dotenv
```

7. Key Scientific Literature & Academic Benchmarks

- **Dubarry et al. — *Incremental Capacity Analysis (ICA):*** Establishes how tracking dQ/dV peak area and position shifts quantifies degradation modes: Loss of Lithium Inventory (LLI) and Loss of Active Material (LAM).
- **Birkl et al. — *Degradation Diagnostics in Li-ion Batteries:*** Formal taxonomy distinguishing mechanical electrode degradation from chemical electrolyte breakdown.
- **SAE J1772 / IEEE Standards:** Industry definitions for standard automotive EV end-of-life criteria (typically 70%–80% capacity retention) and safe DC fast-charging thermal windows.

8. Six-Week Implementation Roadmap

Timeline	Milestone Target	Weekly Work Distribution & Deliverables
Week 1	Data Acquisition & Interface Design	M1 downloads NASA & BatteryArchive data; M2 defines feature schemas; M3 creates vehicle dynamics equations; M4 sets up LangGraph state skeleton; M5 initializes GitHub repo & Streamlit shell.
Week 2	Cleaning Pipelines & Feature Extraction	M1 exports normalized Parquet files; M2 implements dQ/dV filter and thermal exposure counters; M3 validates speed-power consumption curves; M5 integrates raw data plots into UI.
Week 3	ML Model Training & Range Engine	M2 trains XGBoost & LSTM models (MAE < 2%); M3 links SOH output to the 650 km → 550 km vehicle range decay engine; M4 begins AutoGen driver agent simulation scripts.
Week 4	Multi-Agent Simulation & Advisory	M4 links simulated driving stress cycles to ML model to evaluate degradation; M5 builds ChromaDB RAG pipeline indexing solid-state and sodium-ion literature.

Timeline	Milestone Target	Weekly Work Distribution & Deliverables
Week 5	Full Pipeline Integration & UI	M5 connects all 5 modules into the interactive Streamlit dashboard; team stress-tests edge-case temperatures (−10°C to 45°C) and fast-charge profiles across all 4 chemistries.
Week 6	Evaluation, Report & Demo	Export comparative performance charts across chemistries; finalize code documentation, presentation slides, architecture diagrams, and comprehensive final project report.